Assessing Variation in Built Environment Factors and Heat Adaptation in San Francisco

Alani Leonard[1], Matt Holmes[2], Melina Rubio[2 3], Marcelina Lobato[3], Amethyst Blumenshine[2], Mahal Leia Bilaoen BS[3], Nancy Carmona PhD[2]

Department of Biology[1], Department of Public Health[2], The School of Environment[3]

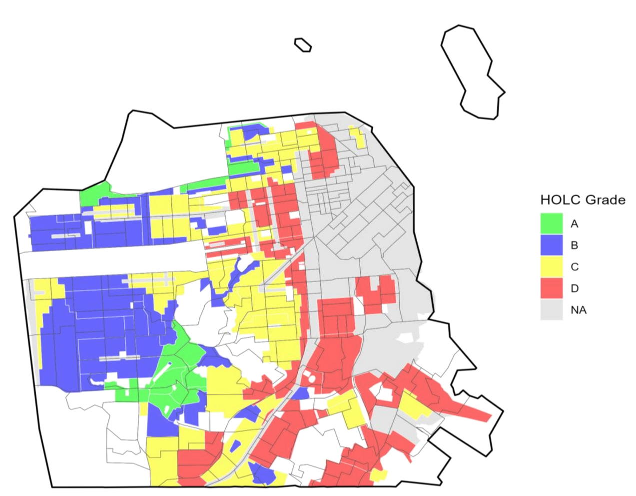
Introduction

- **Rising extreme heat conditions** are an increasing threat to **urban residential health**. [1, 2, 3]
- **Cooling investments** reduce **heat-related negative health outcomes**. [2, 3]
- Higher **Social Vulnerability Index (SVI)** and **lower historical redlining grades** are linked to **lower cooling-related infrastructure investments.** [4,5]

Aim: Examine geographic inequities in heat exposure and cooling investment across San Francisco, California in relation to historical redlining.

Historical Home Owners' Loan Corporation (HOLC) Redlining Grades, 1930s

HOLC grades ranged from A: "Best" to D: "Hazardous"



Methods

- Census tract-year ecological analysis of San Francisco from 2013–2022.[6]
- Calculated apparent temperature (AT) using Daymet daily temperature and vapor pressure using **Steadman's formula**; heatwaves defined as **2+ consecutive days > tract-specific 95th percentile of 1982–2012 AT**.[1, 2, 7]
- Cooling updates defined as residential permits for air conditioning and ventilation.[8, 9, 10]
- Analyses: Descriptive statistics, Kruskal–Wallis and Wilcoxon rank-sum tests, Spearman correlations, and logistic regression.

Table 1. Data Sources

Data source	Key features used	Years	Analysis unit
Daymet	Apparent temp; heatwave events	1982–2022	Census tract-year
SF Building Permits + Assessor Tax Rolls	Cool improvements; year built/age	2013–2022	Census tract-year
SF Tree Inventory Planting Data	Trees planted (proxy for greening invest.)	2013–2022	Census tract-year
Social Vulnerability Index (SVI) from CDC	Overall SVI + four domain scores	2022	Census tract
Mapping Inequality (Home Owner's Loan Corporation)	Historical HOLC redline grade (A–D); primary tract grade	Historical (1930s)	Census tract

Results

- Grade A tracts had the highest median tree planting activity, with **8 trees per tract-year** compared with **3–4 trees** in Grades B–D (Table 2).
- Higher-heat tract-years had more tree planting and cooling-related building updates than lower-heat tract-years ($p < 0.001$ and $p = 0.002$).
- Compared with Grade A tracts, Grades B–D had **47%–56% lower adjusted odds** of cooling-related building updates (Figure 4).

Figure 1. Heat Exposure

- Mean annual number of 2-day heatwave events, 2013–2022.[2]
- Higher values indicate greater heat exposure.
- Exposure did not differ significantly by HOLC grade ($p = 0.50$).

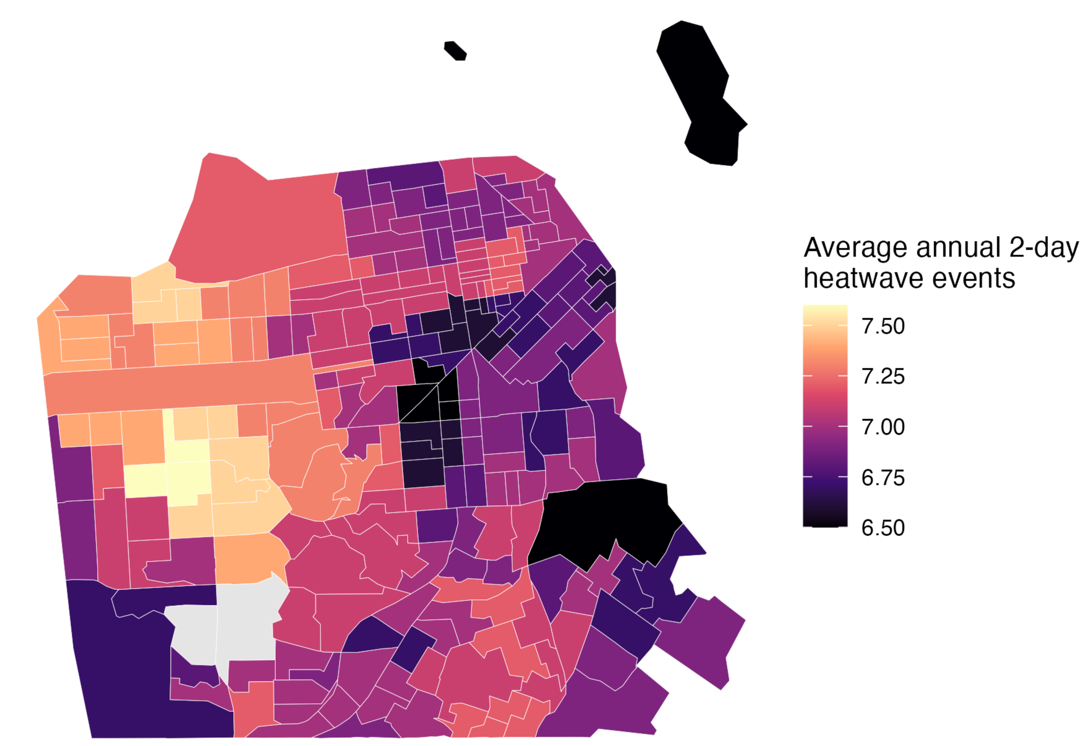


Figure 2. Residential Cooling Updates

- Total residential cooling-related permits by census tract, 2013–2022.[8, 9, 10]
- Includes permitted air-conditioning, ventilation, fan, and related air-system work.
- Higher counts indicate more cooling-related building activity
- Grades B–D had 47%–56% lower adjusted odds of any update than Grade A.

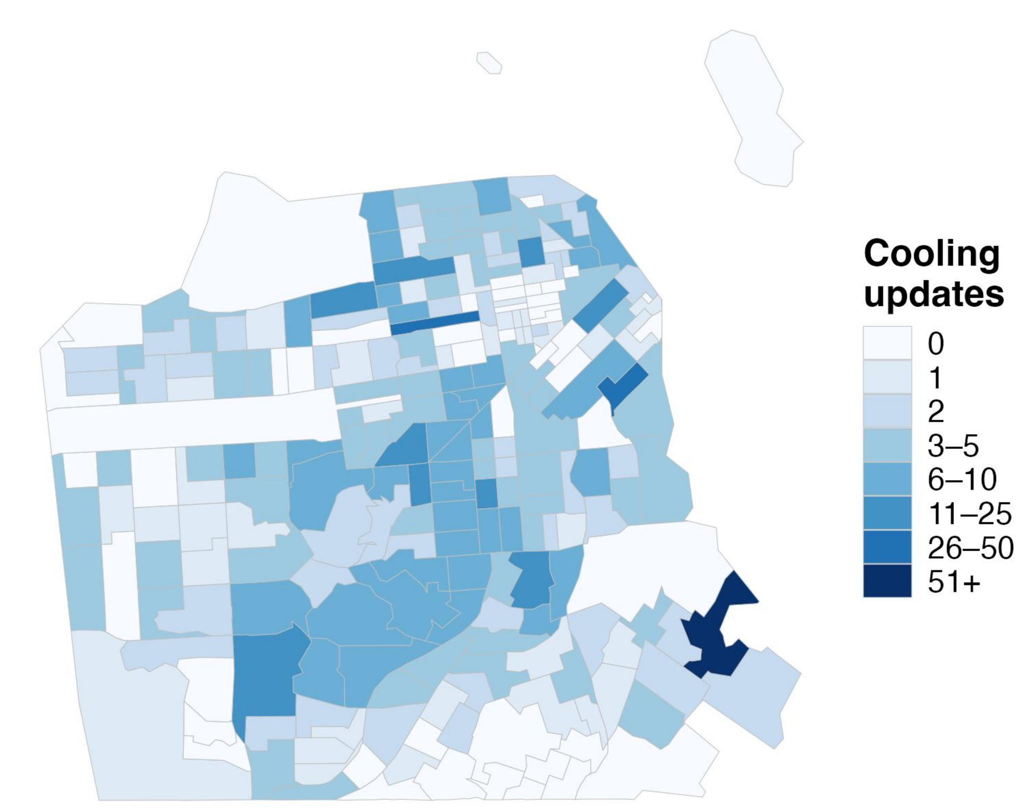


Figure 3. Tree Planting

- Trees planted per census tract-year, 2013–2022.[11]
- Used as a proxy for neighborhood greening investment.
- Grade A tracts had the highest median planting activity: 8 trees versus 3–4 in Grades B–D.
- Highest tree planting Mission, followed by Bayview Hunters Point and Sunset/Parkside.

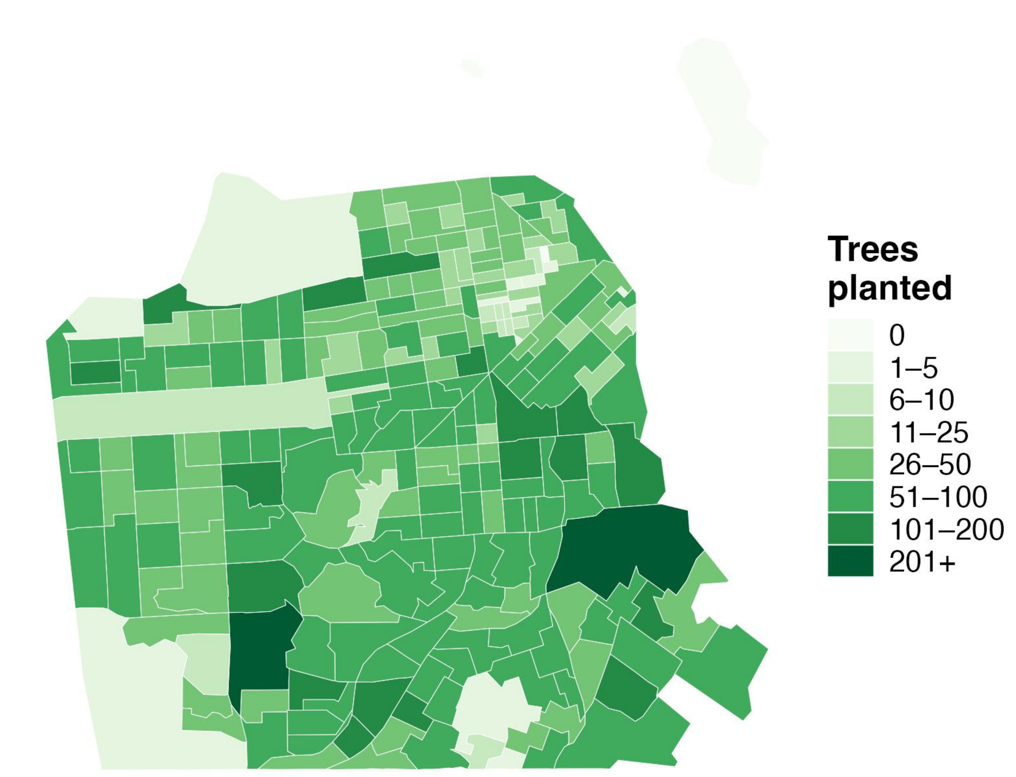
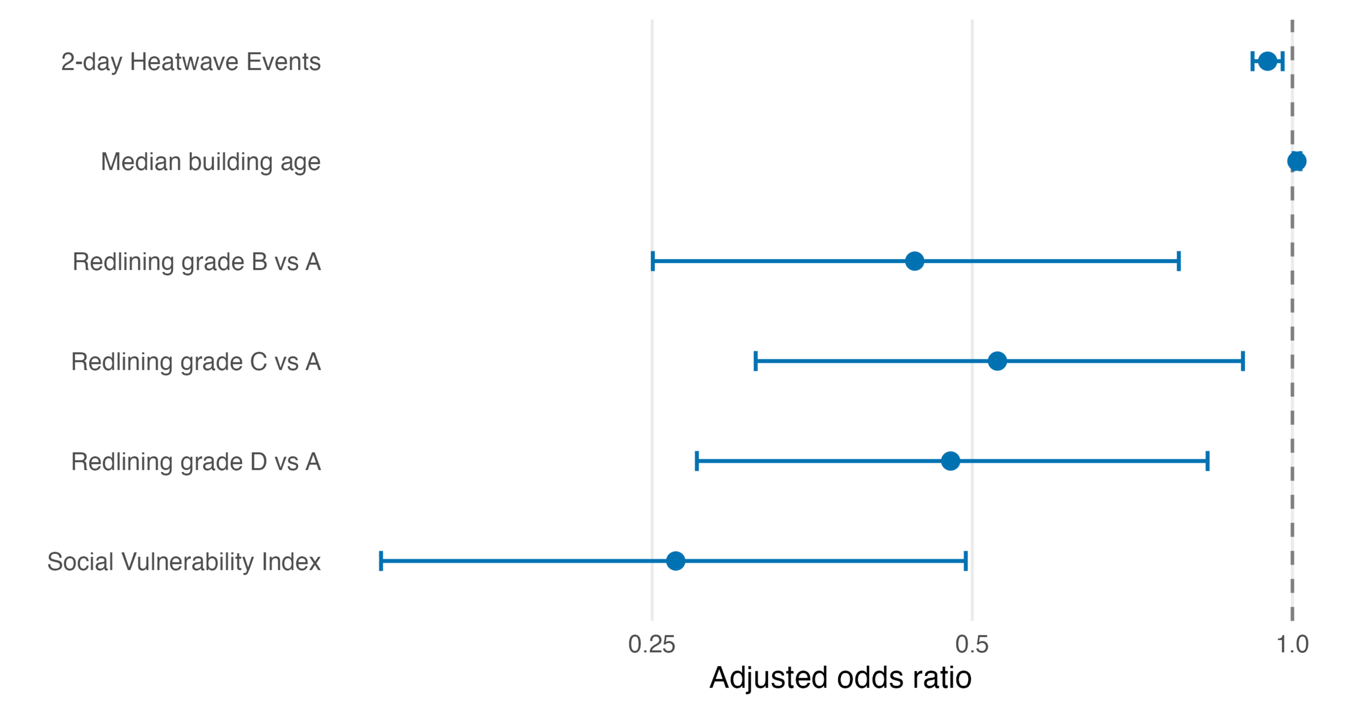


Table 2. Descriptive Characteristics by HOLC Grade

Characteristic, median (IQR)	A: Best N = 67	B: Still desirable N = 336	C: Definitely declining N = 600	D: Hazardous N = 610	p-value
Social Vulnerability Index	0.10 (0.16)	0.37 (0.23)	0.17 (0.36)	0.45 (0.35)	<0.001
2-day heatwave events	8.0 (6.0)	7.0 (7.0)	7.0 (7.0)	7.0 (5.0)	0.478
Cooling updates (%)	0.0 (0.4)	0.0 (0.0)	0.0 (0.4)	0.0 (0.0)	0.001
Trees planted	8 (14)	3 (6)	4 (6)	3 (7)	<0.001
Median building age	99 (16)	90 (14)	102 (19)	105 (35)	<0.001

Note: Values presented as median (IQR); p-values calculated using Kruskal-Wallis rank sum test.

Figure 4. Predictors of Cooling Updates



Conclusion & Implications

- Heat-adaptation resources were more common in hotter census tract-years, suggesting **some investment is reaching areas with greater heat exposure.**
- Cooling-related updates were less likely in historically lower-graded HOLC neighborhoods, indicating **persistent geographic inequities.**
- Future investments should be equitable and prioritize neighborhoods with high heat exposure, greater social vulnerability, and fewer existing cooling resources.

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Citations

1. Steadman, R. G. *A Universal Scale of Apparent Temperature. J. Appl. Meteorol.* **1984**, *23* (12), 1674–1687.
2. Cuce, P. M.; Cuce, E.; Santamouris, M. *Towards Sustainable and Climate-Resilient Cities: Mitigating Urban Heat Islands Through Green Infrastructure. Sustainability* **2025**, *17* (3), 1303. https://doi.org/10.3390/su17031303
3. World Health Organization. *Heat and Health.* https://www.who.int/news-room/fact-sheets/detail/climate-change-heat-and-health
4. Centers for Disease Control and Prevention; Agency for Toxic Substances and Disease Registry. *CDC/ATSDR Social Vulnerability Index 2022 Database: California.*https://www.atsdr.cdc.gov/placeandhealth/svi/data_documentation_download.html
5. Nelson, R. K.; Winling, L.; Marciano, R.; et al. *Mapping Inequality: Redlining in New Deal America.* University of Richmond. https://dsl.richmond.edu/panorama/redlining/
6. DataSF. *Analysis Neighborhoods—2020 Census Tracts Assigned to Neighborhoods.* 2025. https://data.sfgov.org/Geographic-Locations-and-Boundaries/Analysis-Neighborhoods-2020-census-tracts-assigned/sevw-6tgi/about_data
7. Thornton, M. M.; Shrestha, R.; Wei, Y.; Thornton, P. E.; Kao, S.-C. *Daymet: Daily Surface Weather Data on a 1-km Grid for North America, Version 4 R1*; ORNL DAAC: Oak Ridge, TN, 2022. https://doi.org/10.3334/ORNLDAAC/2129
8. DataSF. *Building Permits Before January 1, 2013.* https://data.sfgov.org/
9. DataSF. *Building Permits Filed on or After January 1, 2013.* https://data.sfgov.org/
10. DataSF. *Assessor Historical Secured Property Tax Roll.* https://data.sfgov.org/Housing-and-Buildings/Assessor-Historical-Secured-Property-Tax-Rolls/wv5m-vpq2/about_data
11. DataSF. *Street Tree List.* https://data.sfgov.org/